

JACKSONVILLE, IL, USA

WMO: 744666

Lat: 39.780N

Lon: 90.238W

Elev: 190

StdP: 99.06

Time Zone: -6.00 (NAC)

Period: 02-19

WBAN: 53944

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-17.1	-13.4	-21.3	0.6	-15.3	-18.8	0.7	-12.7	12.4	-4.6	11.2	-2.6	3.2	280	0.494

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.6	33.6	24.6	32.5	24.1	31.2	23.6	26.5	31.4	25.5	30.6	24.7	29.6	3.7	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
25.1	20.7	29.7	24.0	19.3	28.7	23.0	18.2	27.6	83.5	31.3	79.6	30.7	75.8	29.9	29.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
10.6	9.0	8.1	-20.6	35.8	3.1	2.0	-22.9	37.2	-24.7	38.4	-26.4	39.5	-28.7	40.9
			-21.1	27.8	3.0	0.8	-23.2	28.4	-25.0	28.9	-26.7	29.3	-28.9	29.9
			DB											
			WB											

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	12.2	-2.1	-0.5	6.4	12.4	18.3	22.9	24.3	23.5	20.0	13.0	6.5	0.9		
	DBStd	10.69	6.81	6.50	6.39	5.39	4.91	3.32	3.11	3.09	4.26	5.49	5.72	5.63		
	HDD10.0	1317	378	299	151	34	3	0	0	0	1	30	134	287		
	HDD18.3	2903	634	528	374	187	64	4	1	1	30	183	356	542		
	CDD10.0	2117	3	4	38	108	260	387	444	417	299	123	29	3		
	CDD18.3	662	0	0	3	11	63	141	187	160	79	18	1	0		
	CDH23.3	6361	0	0	22	127	548	1314	1916	1468	796	167	4	0		
	CDH26.7	2295	0	0	2	27	157	476	764	539	286	44	0	0		
(14) Wind	WSAvg	3.6	4.3	4.3	4.5	4.5	3.8	3.1	2.4	2.3	2.6	3.3	4.0	4.0		
(15) Precipitation	PrecAvg	980	38	39	77	101	122	109	101	90	95	73	78	57		
	PrecMax	1525	122	123	164	253	268	296	270	198	406	212	277	245		
	PrecMin	646	1	0	15	15	23	16	26	16	0	7	8	6		
	PrecStd	225	29	26	36	50	63	65	66	48	73	43	55	44		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	17.4	19.6	26.2	30.0	32.3	34.0	36.4	35.8	34.1	30.8	23.4	18.0		
		MCWB	13.4	13.0	17.6	18.3	21.9	24.2	24.5	25.0	23.4	21.1	15.2	15.3		
	2%	DB	12.9	15.7	22.4	26.8	30.2	32.6	33.4	33.0	32.2	27.6	20.4	14.0		
		MCWB	10.4	11.5	15.8	16.6	21.1	23.7	25.7	24.4	22.7	18.9	14.6	11.6		
	5%	DB	10.0	12.2	19.3	24.4	28.5	31.3	32.3	31.6	30.2	25.0	17.8	11.6		
		MCWB	7.3	9.0	14.2	16.1	20.1	23.2	25.2	24.3	21.9	17.9	13.4	9.4		
	10%	DB	7.0	8.8	16.4	22.1	26.6	29.7	30.9	29.9	27.8	22.3	15.3	9.1		
		MCWB	4.7	5.8	12.6	14.8	19.0	22.4	24.4	23.5	20.4	16.8	11.5	7.1		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	14.6	15.1	18.6	19.9	24.0	26.2	28.1	27.5	25.1	23.0	17.6	16.1		
		MCDB	16.7	18.0	23.7	26.4	29.9	31.4	32.2	32.2	30.9	29.3	21.3	18.0		
	2%	WB	11.2	11.9	16.6	18.6	22.6	25.2	26.8	26.2	24.0	21.5	15.8	12.4		
		MCDB	12.6	14.4	21.3	24.1	27.8	30.5	31.4	31.2	29.4	25.5	19.0	13.8		
	5%	WB	7.7	8.9	15.1	17.4	21.5	24.3	25.8	25.2	23.1	19.8	13.9	9.6		
		MCDB	9.7	12.2	18.6	22.4	26.6	29.7	30.7	30.2	28.0	22.7	17.2	11.0		
	10%	WB	4.6	5.9	12.7	15.8	20.3	23.2	25.1	24.2	22.1	17.8	11.8	7.1		
		MCDB	7.0	8.7	15.9	20.7	25.0	28.2	29.9	28.3	26.4	21.3	14.8	9.0		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.6	9.1	10.3	11.9	10.8	10.6	10.6	10.8	12.7	11.6	9.8	7.8		
		MCDBR	12.6	13.6	13.7	15.1	12.3	11.7	11.1	12.1	13.7	14.2	12.9	11.1		
	5% WB	MCWBR	10.0	10.1	8.3	7.7	5.5	5.2	5.1	5.3	5.7	7.0	8.4	9.1		
		MCWBR	11.9	13.0	12.3	12.7	10.7	10.4	10.1	10.2	10.8	11.1	11.5	10.6		
(40) Clear-Sky Solar Irradiance	taub		0.295	0.307	0.338	0.388	0.425	0.429	0.455	0.445	0.403	0.351	0.303	0.291		
	taud		2.461	2.421	2.415	2.274	2.233	2.264	2.175	2.209	2.294	2.444	2.524	2.488		
	Ebn at Noon		878	915	917	886	857	851	824	823	836	851	864	855		
	Edh at Noon		78	94	106	130	139	135	146	137	118	92	74	71		
(44) All-Sky Solar Radiation	RadAvg		1.91	2.75	3.71	4.84	5.52	6.29	6.27	5.51	4.73	3.26	2.19	1.61		
	RadStd		0.22	0.26	0.28	0.33	0.43	0.52	0.39	0.34	0.32	0.36	0.20	0.20		

Historical Trends

		DBAvg		Heating		Cooling		Degree-Days			
				99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (35 neighbors)		N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+105	+75

Nomenclature: See separate page